

25 Nov 2015

NEGF is a very useful tool for many part of condensed matter physics. Students of this group should know this theory / formalism very well in order to do good work. This note is intended to give a solid foundation ~~at basic~~ of NEGF.

We begin with quantum-mechanics/Schrödinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = H \psi$$

This is an equation for one single electron, e.g. In such case, it is computationally useful to discretize the wave function (in some basis)

$$\psi = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$$

tight-binding

and write the Schrödinger equation in the $\wedge$ form

$$i\hbar \frac{d}{dt} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} & \cdots & H_{1N} \\ & \ddots & & \\ & & \ddots & \\ H_{N1} & & & H_{NN} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$$

or briefly just $i\hbar \frac{d}{dt} C = H C$
 where C is a column vector of complex numbers.
 H is the single particle Hamiltonian in a matrix form.

For many electron systems it is more convenient to work in occupation number representation and consider 2nd quantized Hamiltonian

$$\hat{H} = \hat{C}^\dagger H \hat{C}$$

here H is the same matrix as before $\hat{C}$ is annihilation operator.

We drop the hat $\wedge$ on c

$$\hat{H} = C^\dagger H C$$

and $C = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$ is column of

annihilation operators and

row of creation operators. $C^\dagger = (c_1^\dagger c_2^\dagger \dots c_N^\dagger)$ is

The Heisenberg equation is

$$i\hbar \frac{dc}{dt} = Hc$$

This has exactly the same form as the Schrödinger equation on p. 3.

□ Exercise: Verify that $i\hbar \frac{dc}{dt} = [c, \hat{H}]$ produces the Schrödinger eq. Hint. Need use $c_i c_j^\dagger + c_j^\dagger c_i = 1$

$$c_i^2 = 0 \text{ etc. } [A, B] \equiv AB - BA$$

Now, we consider equilibrium non-equilibrium Green's functions. Equilibrium means

$$\langle \dots \rangle_{eq} \equiv \text{Tr} \left(\frac{e^{-\beta(H - \mu N)}}{Z} \dots \right) = \text{Tr}(\rho_{eq} \dots)$$

that is grand-canonical distribution where

$$Z = \text{Tr}(e^{-\beta(H - \mu N)} \dots)$$

$\text{Tr}(\dots)$ is over the Fock space, that is

the eigen states of operator $\hat{n}_i = c_i^\dagger c_i \quad i=1, 2, \dots, N$

$|n_1, n_2, \dots, n_N\rangle$ Since each site (or state) can be either occupied $n_i=1$ or empty, we have exactly 2^N possible states. We trace over all those 2^N states in taking the averages $\langle \dots \rangle_{eq}$.

In nonequilibrium open quantum systems, we don't have a nice formula for ρ , at best we can compute reduced density matrix. With the except of Hershfield density matrix $e^{-\beta(H-\psi)}$ but this is purely formal.

We define several versions of Green's functions. In equilibrium, we only need to know one, as they are all related. But it is worthwhile to remember all the definitions. There is no excuse that you don't know the definition t : Heisenberg evolution

Greater: $G_{jk}^>(t, t') = -\frac{i}{\hbar} \langle C_j(t) C_k^\dagger(t') \rangle$
 create 1st

since $i \equiv \text{Fl}$. I try avoid use i for the site/state.

Lesser: $G_{jk}^<(t, t') = +\frac{i}{\hbar} \langle C_k^\dagger(t') C_j(t) \rangle$

since we swap a fermi operator we need + sign here. For Boson operators we do not have sign change

$G^>$ is a matrix, so we can also write

$$G^>(t, t') = -\frac{i}{\hbar} \langle C(t) C^\dagger(t') \rangle$$

$$= -\frac{i}{\hbar} \langle \begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{pmatrix} (C_1^\dagger C_2^\dagger \dots C_n^\dagger) \rangle$$

$G^<$ can also be written in this compact form but it is a bit tricky. Please check that it is

$$G^<(t, t') = \frac{i}{\hbar} \left(\langle C^\dagger(t') C(t)^T \rangle \right)^T$$

But this is ugly, we'll not use it.

The retarded is most useful in a sense, since it is related to linear response theory.

$$G^r(t, t') = \theta(t - t') (G^>(t, t') - G^<(t, t'))$$

or $G^r_{jk}(t, t') = \frac{-i}{\hbar} \theta(t - t') \langle [C_j(t), C_k^\dagger(t')] \rangle$

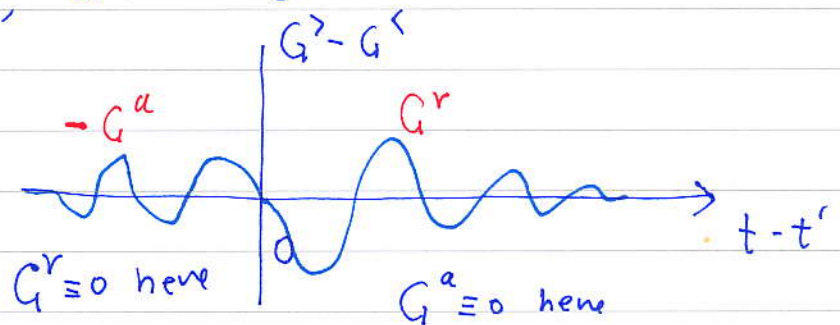
For boson, we use commutator. $\{A, B\} = AB + BA$ is anti-commutator.

$$G^a(t, t') = -\theta(t' - t) (G^>(t, t') - G^<(t, t'))$$

Let define $G^> - G^< = G^r - G^a = -iA$

Since $\theta(t - t') + \theta(t' - t) = 1$

G^r is the positive half, G^a is negative half with a sign flipped.



A is an important quantity. Its Fourier transform is $2\pi \times$ density of states. A is the spectrum function.

Equation of motion

We compute $i\hbar \frac{\partial}{\partial t}$ to the Green's function and find

$$(i\hbar \frac{\partial}{\partial t} - H) G^{><}(t, t') = 0$$

$$(i\hbar \frac{\partial}{\partial t} - H) G^r(t, t') = +\delta(t - t') I$$

They are the Green's function for the Schrödinger equation in the classical physics/math sense.

Let prove the 2nd Equation

$$G_{jk}^r(t, t') = -\frac{i}{\hbar} \theta(t-t') \langle C_j(t) C_k^\dagger(t') + C_k^\dagger(t') C_j(t) \rangle$$

differentiate wth resps to t

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} G_{jk}^r(t, t') &= i\hbar \left(-\frac{i}{\hbar} \right) \left[\frac{\partial}{\partial t} \theta(t-t') \right] \langle C_j(t) C_k^\dagger(t') + C_k^\dagger(t') C_j(t) \rangle \\ &\quad + \left(-\frac{i}{\hbar} \right) \langle i\hbar \frac{\partial}{\partial t} C_j(t) C_k^\dagger(t') + C_k^\dagger(t') i\hbar \frac{\partial}{\partial t} C_j(t) \rangle \\ &= \delta(t-t') \langle C_j(t) C_k^\dagger(t') + C_k^\dagger(t') C_j(t) \rangle \\ &\quad + \left(-\frac{i}{\hbar} \right) \langle \sum_l H_{jl} C_l(t) C_k^\dagger(t') + \sum_l C_k^\dagger(t') H_{jl} C_j(t) \rangle \end{aligned}$$

We have used the fact that $\frac{\partial}{\partial t} \theta(t-t') = \delta(t-t')$

and $i\hbar \frac{dC}{dt} = HC$

due to the δ -fnctn in the 1st term we can set $t' = t$

but $\{C_j, C_k^\dagger\} = \delta_{jk}$ at equal time

$$\text{so } i\hbar \frac{\partial}{\partial t} G_{jk}^r(t, t') = \delta(t-t') \delta_{jk} + \sum_l H_{jl} G_{lk}^r(t, t')$$

In matrix form $i\hbar \frac{\partial}{\partial t} G^r(t, t') = \delta(t-t') I + H G^r(t, t')$

$$\text{or } i\hbar \frac{\partial}{\partial t} G^r(t, t') - H G^r(t, t') = \delta(t-t') I.$$

We can solve the retarded Green's function easily if we go to energy space (i.e. frequency domain) we define

$$\tilde{G}(E) \equiv \int_{-\infty}^{\infty} dt e^{i\frac{E}{\hbar}t} G(t)$$

The inverse transform is $G(t) = \int_{-\infty}^{+\infty} \frac{dE}{2\pi\hbar} e^{-i\frac{E}{\hbar}t} \tilde{G}(E)$
 we can also identify $\omega = E/\hbar$
 but I prefer to use E . Moreover

we have
$$\delta(t) = \int \frac{dE}{2\pi\hbar} e^{-i\frac{E}{\hbar}t} \cdot 1$$

It is clearly $G^r(t, t') \equiv G^r(t-t')$ in practice
 equilibrium Green's function is always time translationally
 invariant. we have only one argument

$$i\hbar \frac{\partial}{\partial t} G^r(t) - H G^r(t) = \delta(t)$$

become. in E -space

$$\left\{ (i\hbar) \left(-i\frac{E}{\hbar}\right) I - H \right\} \tilde{G} = 1$$

$$I = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

N -dimensional
identity
matrix

or
$$\tilde{G}^r(E) = (E I - H)^{-1}$$

However, this form is not well defined when we do
 inverse Fourier transform. we need to shift the
 poles to lower part (negative imaginary) so
 the correct formula is

$$\tilde{G}^r(E) = (E + i\eta) I - H)^{-1}$$

$$\eta \rightarrow 0^+$$

If we use the same method to $G^<$, we get

$$(E \cdot I - H) \tilde{G}^< = 0$$

Does this mean $\tilde{G}^< = 0$? Of course not.

Unless $E \cdot I - H \stackrel{?}{=} 0$

$$f(E) = \frac{1}{e^{\beta(E-\mu)} + 1}$$

Formally we can write

$$\tilde{G}(E) = \text{const.} \delta(E - H)$$

In fact, this is correct except we cannot fix the constant. The correct expression is

$$\tilde{G}^<(E) \underset{\text{min sign here}}{=} -f(E) (\tilde{G}^r(E) - \tilde{G}^a(E)) = i f(E) \tilde{A}(E)$$

We prove this by different means. This is an important relation in equilibrium. It is called "fluctuation dissipation theorem". This relation does not hold in nonequilibrium as there is no fermi distribution f to talk about. Fortunately, there is a replacement to it. That is the Keldysh equation. So in summary

- For equilibrium systems we have fluctuation-dissipation:

$$\tilde{G}^< = -f(\tilde{G}^r - \tilde{G}^a) = i f \tilde{A}$$

- For nonequilibrium case, we get Keldysh equation

$$\tilde{G}^< = \tilde{G}^r \tilde{\Sigma}^< \tilde{G}^a$$

where $\Sigma^<$ is self-energy.

The tilde $\sim$ means we are in energy domain!

Note that both equations are in energy domain

$$\tilde{G} \rightarrow \tilde{G}(E) \quad \text{not in time domain } G(t).$$

The Keldysh equation is perhaps the most important equation in NEGF. It is a consequence of the contour variable Dyson equation.

We prove a very general fluctuation-dissipation theorem. We define

$$G_{AB}^>(t) = -\frac{i}{\hbar} \langle A(t) B(0) \rangle$$

$$G_{AB}^r(t) = -\frac{i}{\hbar} \theta(t) \langle A(t) B \pm B A(t) \rangle$$

$$= \theta(t) (G_{AB}^>(t) - G_{AB}^<(t))$$

$$-i A_{AB}(t) = -\frac{i}{\hbar} \langle A(t) B \pm B A(t) \rangle = G_{AB}^>(t) - G_{AB}^<(t)$$

$$= G_{AB}^r(t) - G_{AB}^a(t)$$

These definitions are consistent to those in page 7, 9, except that A and B are arbitrary quantum operators, hermitian or not. Now the $\pm$ sign. If A, B are fermion-like we take $+$, boson-like we take $-$. Think of $\frac{1}{e^{\beta(\epsilon \pm \mu)} \pm 1}$ $\leftarrow$ $+$ fermi distribution
 $-$ bose distribution.

Fermion-like means the operator follows anti-commutation relation like c and $c^\dagger$. Boson-like means the operator follows commutation relation. More precisely, under the time order or contour order sign T_τ , boson operators commute, fermion operators anti-commute. Even though we are dealing with fermions (electrons), we still have boson-like operators like local energy, current j , or charge ρ as they are pair of fermion operators like $c^\dagger c$.

The idea of proof is to use Lehmann representation i.e. expand the --- expression of Green's functions in the eigen state of $\hat{H}$ and then use the

Plemelj formula $\frac{1}{x \pm i\eta} = P \frac{1}{x} - i\pi \delta(x)$ where

$P \frac{1}{x}$ stand for principle value

□ Question. what is a "principle value"?

Let $\hat{H}|\mu\rangle = E_\mu|\mu\rangle \quad \mu=1,2,\dots, 2^N$

This is a "many-body" state. in Fock space. Since the dimension of the Fock space is 2^N , we have 2^N different states denoted by $|\mu\rangle$. This space is much larger than the single particles of eigen vector of H , where

$\hat{H} = C^\dagger H C$. The dimension of single particle state is

N . For such non-interacting system, standard procedure exists to build $|\mu\rangle$ from the single particle state ϕ_i , where $H\phi_i = \epsilon_i\phi_i$ where ϵ

is single particle energy. We leave this as an exercise.

□ Question: How to build $|\mu\rangle$ from ϕ_i ?

We'll work exclusively in $|\mu\rangle$, the many-body state. We work for $G^>$ where Heisenberg operator

$$A(t) = e^{\frac{i}{\hbar} \hat{H} t} A e^{-\frac{i}{\hbar} \hat{H} t}$$

$$B(\omega) = B$$

$$G^>(t) = \frac{1}{i\hbar} \langle A(t) B(\omega) \rangle = \frac{1}{i\hbar} \text{Tr} \left(\frac{e^{-\beta(\hat{H} - \mu \hat{N})}}{Z} e^{\frac{i}{\hbar} \hat{H} t} A e^{-\frac{i}{\hbar} \hat{H} t} B \right)$$

We use the fact that $\hat{H}$ and $\hat{N}$ commute, and so it is always possible that $\hat{N}|\mu\rangle = N_\mu|\mu\rangle$ is also an eigen state of $\hat{N}$.

Trace means $\sum_{\mu} \langle \mu | \dots | \mu \rangle = \text{Tr}(\dots)$

$$G_{AB}^>(t) = \frac{1}{i\hbar} \sum_{\mu} \langle \mu | \frac{1}{Z} e^{-\beta(\hat{H} - \mu \hat{N})} e^{\frac{i}{\hbar} \hat{H} t} A e^{-\frac{i}{\hbar} \hat{H} t} B | \mu \rangle$$

$\sum_{\nu} |\nu\rangle \langle \nu| = \hat{1}$ insert complete set

$$= \frac{1}{i\hbar Z} \sum_{\mu, \nu} e^{-\beta(E_{\mu} - \mu N_{\mu})} \dots$$

sorry for notation
confusion let use n & m instead

$$= \frac{1}{i\hbar Z} \sum_{n, m} e^{-\beta(E_n - \mu N_n) + \frac{i}{\hbar} E_n t} \langle n | A | m \rangle \langle m | B | n \rangle$$

Go to energy space by Fourier transform

$$\tilde{G}_{AB}^>(E) = \int_{-\infty}^{+\infty} dt G_{AB}^>(t) e^{\frac{iEt}{\hbar}} = \frac{1}{i\hbar Z} \sum_{n, m} \langle n | A | m \rangle \langle m | B | n \rangle \left\{ \begin{array}{l} E + \\ 2\pi \delta\left(\frac{E_n - E_m}{\hbar}\right) \cdot e^{-\beta(E_n - \mu N_n)} \end{array} \right\}$$

$$= \frac{1}{iZ} \sum_{n, m} \langle n | A | m \rangle \langle m | B | n \rangle e^{-\beta(E_n - \mu N_n)} \cdot 2\pi \delta(E_n - E_m + E)$$

we have use the formula $\int_{-\infty}^{+\infty} e^{ixt} dt = 2\pi \delta(x)$

notice the 2π here

and $\delta(ax) = \frac{1}{|a|} \delta(x)$.

$G^>$ is done, now let focus on $A_{AB}(t)$

$$A_{AB}(t) = i(G_{AB}^>(t) - G_{AB}^<(t))$$

$$= \frac{1}{\hbar} \langle \underbrace{A(t)B}_{\text{already worked out}} \pm B A(t) \rangle$$

swap pos. of A & B .

The Fourier transform of A_{AB} is

$$\begin{aligned}
 \tilde{A}_{AB}(E) &= \frac{1}{Z} \sum_{n,m} \left\{ e^{-\beta(E_n - \mu N_n)} \cdot 2\pi \delta(E + E_n - E_m) \langle n|A|m\rangle \langle m|B|n\rangle \right. \\
 &\quad \left. \pm e^{-\beta(E_n - \mu N_n)} 2\pi \delta(E + E_m - E_n) \langle n|B|m\rangle \langle m|A|n\rangle \right\} \\
 &= \frac{1}{Z} \sum_{n,m} \left\{ \left(e^{-\beta(E_n - \mu N_n)} \pm e^{-\beta(E_m - \mu N_m)} \right) 2\pi \delta(E + E_n - E_m) \langle n|A|m\rangle \langle m|B|n\rangle \right\}
 \end{aligned}$$

swap index $n \leftrightarrow m$

Due to the δ -function constraints, we have $E + E_n - E_m = 0$
 we have $E_m = E + E_n$

Also if $\langle n|A|m\rangle \neq 0$ $N_n = N_m$, + transition between different N is possible if A is c or $c^\dagger$. We need consider two cases. if A or B is of the form $c^\dagger c$ then $N_n = N_m$. if $A = c$, $B = c^\dagger$

then $N_n = N_m - 1$ in order to have non zero $\langle n|A|m\rangle$
 $\langle m|B|n\rangle$

For Bosonic case (A or B of the form $c^\dagger \dots c$)

$$\begin{aligned}
 &= \frac{1}{Z} \sum_{n,m} \left(e^{-\beta(E_n - \mu N_n)} - e^{-\beta(E + E_n - \mu N_n)} \right) \cdot 2\pi \delta(\dots) \\
 &= (1 - e^{-\beta E}) \frac{1}{Z} \sum_{n,m} e^{-\beta(E_n - \mu N_n)} \cdot 2\pi \delta(E + E_n - E_m) \langle n|A|m\rangle \langle m|B|n\rangle
 \end{aligned}$$

compare with $\tilde{G}_{AB}^>(E)$ we get

$$N(E) \equiv \frac{1}{e^{\beta E} - 1}$$

$$\tilde{A}_{AB}(E) = (1 - e^{-\beta E}) \cdot i \tilde{G}_{AB}^>(E)$$

$$\begin{aligned}
 \text{or } \tilde{G}_{AB}^>(E) &= \frac{-i}{1 - e^{-\beta E}} \tilde{A}_{AB}(E) = -i \left(\frac{1 - e^{-\beta E} + e^{-\beta E}}{1 - e^{-\beta E}} \right) \tilde{A}_{AB}(E) \\
 &= -i (1 + N(E)) \tilde{A}_{AB}(E)
 \end{aligned}$$

we get, for bosonic operator which can serve particle number (in the sense $A|n\rangle$ and $|n\rangle$ has the same number of electrons). We are dealing with electrons, not real boson particles)

$$\square \quad \tilde{G}^> = -i(1+N)\tilde{A}$$

$$\tilde{G}^> \rightarrow \tilde{G}_{AB}^>(E)$$

$$\tilde{A} \rightarrow \tilde{A}_{AB}(E)$$

since $-i\tilde{A} = \tilde{G}^> - \tilde{G}^<$

$$\begin{aligned} \tilde{G}^< &= \tilde{G}^> + i\tilde{A} = -i(1+N)\tilde{A} + i\tilde{A} \\ &= -iN\tilde{A} = N(E)(\tilde{G}_{AB}^r(E) - \tilde{G}_{AB}^a(E)) \end{aligned}$$

This is a correct result. since $-iA = G^r - G^a$.

$$N(E) = \frac{1}{e^{\beta E} - 1}. \text{ is bose fraction.}$$

Now we consider fermionic case with + sign and

$$N_n = N_m - 1$$

$$\begin{aligned} \text{we get } \tilde{A}_{AB}(E) &= \frac{1}{Z} \sum_{n,m} \left\{ e^{-\beta(E_n - \mu N_n)} + e^{-\beta(E + E_n - \mu(N_n + 1))} \right\} 2\pi \delta(E + E_n - E_m) \langle n|A|m\rangle \langle m|B|n\rangle \\ &= \frac{1 + e^{-\beta(E - \mu)}}{Z} \sum_{n,m} e^{-\beta(E_n - \mu N_n)} 2\pi \delta(E + E_n - E_m) \langle n|A|m\rangle \langle m|B|n\rangle \end{aligned}$$

$$= i(1 + e^{-\beta(E - \mu)}) \tilde{G}_{AB}^>(E)$$

Omitting subscript AB,

$$\tilde{G}^>(E) = \frac{-i}{1 + e^{-\beta(E - \mu)}} \tilde{A}(E)$$

$$= -i(1 - f(E)) \tilde{A}(E) = (1 - f(E))(\tilde{G}^r - \tilde{G}^a)$$

$1 - f$ means the occupation of holes.

Similarly $-iA = G^r - G^a$ and $G^r = (1-f)(-iA)$
 gives $G^a = G^r + iA = (1-f)(-iA) + iA$

fluctuation-dissipation theorem: $= if A = -f(G^r - G^a)$

Or $\tilde{G}_{AB}^<(E) = if(E) \tilde{A}_{AB}(E) = -f(E) (\tilde{G}_{AB}^r(E) - \tilde{G}_{AB}^a(E))$

fermion case has a minus sign here.

$$f(E) \equiv \frac{1}{e^{\beta(E-\mu)} + 1}$$

$$N(E) \equiv \frac{1}{e^{\beta E} - 1}$$

N has no chemical potential μ .

The remain relations $\tilde{A}$ and $\tilde{G}^r$, $\tilde{G}^r$ and $\tilde{G}^a$!

In proving the above relation $G^a = if A = -f(G^r - G^a)$
 we have used the fact the B operator create one particle
 and A operator annihilates one particle. If B operator
 happen to be creating two particles (two electrons), we
 would get $e^{\beta(E-2\mu)}$ instead! In practice
 this equation always mean

$$\tilde{G}_{C_j, C_k^+}^<(E) \quad \text{for } A = C_j, \quad B = C_k^+$$

The most important formula for electron Green's functions.

1) $\tilde{G}^r(E) = \frac{1}{(E + i\eta)I - H}$ ← matrix inverse $\eta \rightarrow 0^+$

2) $\tilde{G}^<(E) = if(E) \tilde{A}(E)$
 $= -f(E) (\tilde{G}^r(E) - \tilde{G}^a(E))$

These two formulas should be memorized/learned by heart.

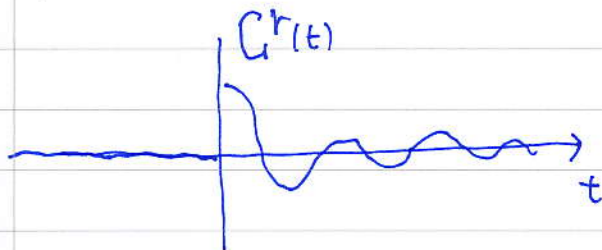
We now discuss the relationship between the spectrum function A and retarded and advanced Green's function G^r and G^a . It is very simple in time domain

$$G^r(t) = \theta(t) (G^>(t) - G^<(t))$$

$$= -i \theta(t) A(t)$$

$$G^a(t) = i \theta(-t) A(t)$$

G^r takes positive t half of A , G^a takes negative half of A with a sign flip.



The feature that

$$G^r(t) \equiv 0 \quad \text{for } t < 0$$

is called causality.

This fact implies that $\tilde{G}^r(E)$ is an analytic function of E in the analytically continuous the retarded Green's function to the v half plane $\mathbb{Z}$. The meaning upper.

of causality is best seen in the linear response theory. (see, eg. N. Potthier, "Nonequilibrium Stat Phys")

If the original Hamiltonian $\hat{H}$ is perturbed by some external perturbation in the form

$$\hat{H}_{tot} = \hat{H} - b(t) \cdot \hat{B}$$

where $b(t)$ is a scalar control field (like classical magnetic or electric field) and $\hat{B}$ is quantum operator then

$$\langle \hat{A}(t) \rangle = - \int_{-\infty}^{\infty} dt' G_{AB}^r(t, t') b(t')$$

Since past can influence the future, but future cannot have effect to the past, we must have

$$G_{AB}^r(t, t') = 0 \quad \text{if} \quad t < t'$$

Under time indepent Hamiltonian $\hat{H}$ which has no explicit time
 $G_{AB}^r(t, t') \equiv G_{AB}^r(t-t')$ depends on the time difference
 only. $t < t'$ means $G_{AB}^r(\Delta t)$ with $\Delta t = t - t' < 0$

Back to relation of A and G^r — since

$$G^r(t) = -i \theta(t) A(t)$$

is a product of two functions $\theta(t)$ and $A(t)$, we can compute $\tilde{G}^r(E)$, the Fourier space retarded Green's function by applying the convolution theorem in reverse: product in time domain becomes convolution in energy domain.

$$A(t) \rightarrow \tilde{A}(E) = \int_{-\infty}^{+\infty} dt e^{i \frac{Et}{\hbar}} A(t)$$

$$\theta(t) \rightarrow \tilde{\theta}(E) = \int_{-\infty}^{+\infty} dt e^{i \frac{Et}{\hbar}} \theta(t) = \int_0^{+\infty} dt e^{i \frac{Et}{\hbar}}$$

Unfortunately the 2nd integral does not exist, and it is also not Lebesgue integral as $\int_0^{+\infty} dt |e^{i \frac{Et}{\hbar}}| \rightarrow +\infty$.

Physicists way to solve this mathematical problem is to add artificial damping. i.e. change the problem to

$$\begin{aligned} \tilde{\theta}_\eta(E) &= \int_0^{+\infty} dt e^{i \frac{Et}{\hbar} - \frac{\eta}{\hbar} t} \quad \eta \rightarrow 0^+ \\ &= \frac{e^{(iE - \frac{\eta}{\hbar})t}}{iE - \frac{\eta}{\hbar}} \Big|_0^{+\infty} = \frac{i\hbar}{E + i\eta} \end{aligned}$$

Now we do the reverse convolution theorem

$$\begin{aligned}
 \tilde{G}^r(E) &= -i \int_{-\infty}^{\infty} dt \, \Theta(t) A(t) e^{i \frac{E}{\hbar} t} \\
 &= -i \int_{-\infty}^{\infty} dt \left[\int_{-\infty}^{\infty} \frac{dE'}{2\pi\hbar} \underbrace{\tilde{\Theta}_\eta(E')}_{\Theta(t)} e^{-i \frac{E'}{\hbar} t} \right] A(t) e^{i \frac{E}{\hbar} t} \\
 &= \int_{-\infty}^{\infty} \frac{dE'}{2\pi\hbar} \tilde{\Theta}_\eta(E') (-i) \int_{-\infty}^{\infty} dt A(t) e^{i \frac{(E-E')}{\hbar} t} = \int_{-\infty}^{\infty} \frac{dE'}{2\pi\hbar} \tilde{\Theta}_\eta(E') \tilde{A}(E-E') \\
 &= \int_{-\infty}^{\infty} \frac{dE'}{2\pi} \cdot \frac{\tilde{A}(E-E')}{E' + i\eta} \quad \text{No } \hbar \text{ gets cancelled} \\
 &= - \int_{+\infty}^{-\infty} \frac{dE''}{2\pi} \cdot \frac{\tilde{A}(E'')}{E - E'' + i\eta} \quad \text{let } E - E' = E'' \\
 &= \int_{-\infty}^{+\infty} \frac{dE'}{2\pi} \frac{\tilde{A}(E')}{E + i\eta - E'} \quad \text{rename } E'' \text{ as } E'
 \end{aligned}$$

This is the final expression for the spectrum represented by $\tilde{G}^r(E)$

□ Question. we get two expressions for $\tilde{G}^r$, one is $\tilde{G}^r(E) = (E + i\eta - H)^{-1}$, and ^{another} $\tilde{G}^r$ is the above spectral integral. What is the difference? Which one more general. How to go from the one form to the other. i.e. their relationship?

□ Show, by the same idea as above that

$$\tilde{G}^a(E) = \int_{-\infty}^{\infty} \frac{dE'}{2\pi} \frac{\tilde{A}(E')}{E - i\eta - E'}$$

↑ everything is the same except a sign change here.

ans: $G^q(t) = i \theta(-t) \tilde{A}(t)$

$\uparrow$ $\uparrow$
 sign differ from $G^r(t)$

$$\theta(-t) = \int_{-\infty}^{\infty} \frac{dE'}{2\pi\hbar} \theta_{\eta}(E') e^{\frac{i}{\hbar} E' t}$$

$\uparrow$ $\uparrow$
 same + instead of -

$$\begin{aligned} \text{So } \tilde{G}^q(E) &= i \int_{-\infty}^{\infty} dt \theta_{\eta}(-t) A(t) e^{i \frac{E}{\hbar} t} \\ &= i \int_{-\infty}^{\infty} \frac{dE'}{2\pi\hbar} \theta_{\eta}(E') \int_{-\infty}^{\infty} dt A(t) e^{i \frac{E+E'}{\hbar} t} \end{aligned}$$

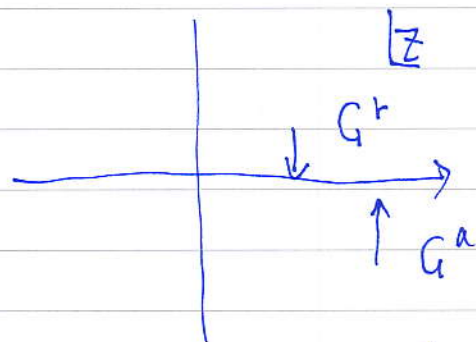
$$= i \int_{-\infty}^{\infty} \frac{dE'}{2\pi\hbar} \frac{i\hbar}{E' + i\eta} \tilde{A}(E+E')$$

let $E+E' = E''$

$$= - \int_{-\infty}^{\infty} \frac{dE''}{2\pi} \frac{\tilde{A}(E'')}{E'' - E + i\eta} = \int_{-\infty}^{\infty} \frac{dE'}{2\pi} \frac{\tilde{A}(E')}{E - E' - i\eta} \quad \text{as claimed}$$

It is useful to define

$$G(z) = \int_{-\infty}^{\infty} \frac{dE'}{2\pi} \frac{\tilde{A}(E')}{z - E'}$$



then $G^r(E) = \lim_{\eta \rightarrow 0^+} G(E + i\eta)$

$$G^a(E) = \lim_{\eta \rightarrow 0^+} G(E - i\eta)$$

If we approach the real axis from above we get G^r , from below

we get $G^a(E)$. The limit is not the same since there is a discontinuity across real axis.

using the Plemelj formula

$$\frac{1}{x + i\eta} = \mathcal{P} \frac{1}{x} - i\pi \delta(x)$$

We can write

$$\tilde{G}^r(E) = P \int_{-\infty}^{+\infty} \frac{dE'}{2\pi} \frac{\tilde{A}(E')}{E - E'} - i \frac{1}{2} \tilde{A}(E)$$

$$\tilde{G}^a(E) = P \int_{-\infty}^{+\infty} \frac{dE'}{2\pi} \frac{\tilde{A}(E')}{E - E'} + i \frac{1}{2} \tilde{A}(E)$$

Subtract, we get

$$i = \sqrt{-1}$$

$$\tilde{G}^r(E) - \tilde{G}^a(E) = -i \tilde{A}(E)$$

This is nothing but the energy domain eqn for the definition of A

$$A(t) = i(G^r(t) - G^a(t))$$

or we write

$$A = i(G^r - G^a) = i(G^> - G^<)$$

which works in t or E space.
as well as

Finally, we need to work out the relationship between $\tilde{G}^r$ and $\tilde{G}^a$. The usual formula is

$$\tilde{G}^a(E) = [\tilde{G}^r(E)]^\dagger \leftarrow \text{hermitian conjugate.}$$

But it is trickier than you think if the operator A and B forming G_{AB}^r is arbitrary operator.

We start from the Lehman representation for $\tilde{G}_{AB}^r(E)$.

$$\text{using } \tilde{G}_{AB}^r(E) = \int_{-\infty}^{+\infty} \frac{dE'}{2\pi} \frac{\tilde{A}_{AB}(E')}{E + i\eta - E'} \quad \text{page 33}$$

$$\text{and } \tilde{A}_{AB}(E') = \frac{1}{Z} \sum_{n,m} \left(e^{-\beta(E_n - \mu N_n)} \pm e^{-\beta(E_m - \mu N_m)} \right) \cdot 2\pi \delta(E' + E_n - E_m) \langle n|A|m\rangle \langle m|B|n\rangle$$

+ fermion-like
- boson-like

We can write

$$\tilde{G}_{AB}^r(E) = \frac{1}{Z} \sum_{n,m} \frac{(e^{-\beta(E_n - \mu N_n)} \pm e^{-\beta(E_m - \mu N_m)})}{E + i\eta - (E_m - E_n)} \langle n | A | m \rangle \langle m | B | n \rangle$$

$\tilde{G}_{AB}^a(E)$ is obtained by replace $i\eta$ by $-i\eta$

Take the complex conjugate of $\tilde{G}_{AB}^r(E)$ we get

$$\begin{aligned} \left(\tilde{G}_{AB}^r(E) \right)^* &= \frac{1}{Z} \sum_{n,m} \frac{\dots}{E - i\eta - E'} \left(\langle n | A | m \rangle \langle m | B | n \rangle \right)^* \\ &= \frac{1}{Z} \sum_{n,m} \frac{(e^{i\beta(E_n - \mu N_n)} \pm e^{i\beta(E_m - \mu N_m)})}{E - i\eta - E'} \langle n | B^\dagger | m \rangle \langle m | A^\dagger | n \rangle \end{aligned}$$

$E' = E_m - E_n$

we have used the genl relation

$$\langle \psi | \hat{O} | \psi \rangle^* = \langle \psi | \hat{O}^\dagger | \psi \rangle$$

Note also $\langle \psi | \psi \rangle^* = \langle \psi | \psi \rangle$

the result is not $\tilde{G}_{BA}^a(E)$ but

$$\boxed{\tilde{G}_{AB}^r(E)^* = \tilde{G}_{B^\dagger A^\dagger}^a(E)}$$

If A and B are hermitian operators then $A^\dagger = A$, $B^\dagger = B$

we get $\tilde{G}^a = (\tilde{G}^r)^\dagger = (\tilde{G}^*)^T$ $\leftarrow$ transpose means swap the index of the matrix

For the case of electron single particle Green's function $B = c_k^\dagger$, $A = c_j$
then $B^\dagger = (c_k^\dagger)^\dagger = c_k$, $A^\dagger = (c_j)^\dagger = c_j^\dagger$, then

$$\tilde{G}_{c_j c_k^\dagger}^{r*}(E) = \tilde{G}_{c_k c_j^\dagger}^a(E) \quad \text{ie.} \quad [\tilde{G}^r]^\dagger = \tilde{G}^a \quad \text{still holds}$$

Matsubara Green's function is needed if we do dynamic mean-field. It is defined as $\langle T_\tau C_j(\tau) C_k^\dagger(\tau') \rangle$ Only Matsubara, contour version has diagrammatic expansions.

$$G_{jk}^M(\tau, \tau') = -\frac{1}{\hbar} \langle T_\tau C_j(\tau) C_k^\dagger(\tau') \rangle$$

where τ, τ' is not contour time but "imaginary time" $i\tau = \tau$.

$$C_j(\tau) \equiv e^{\frac{1}{\hbar} (H - \mu N) \tau} C_j e^{-\frac{1}{\hbar} (H - \mu N) \tau}$$

In Matsubara approach, it is customary to include the chemical potential term $-\mu N$. So even if we take $\tau = i\tau$, it is not Heisenberg operator. The reason for this $-\mu N$ factor is that it makes it the same as distribution eq. $\rho = \frac{e^{-\beta(H - \mu N)}}{\mathcal{Z}}$

The T_τ super operator arranges the order of operator according to the value of τ from small to large, i.e.

$$T_\tau A(\tau) B(\tau') = \begin{cases} A(\tau) B(\tau') & \text{if } \tau > \tau' \\ -B(\tau') A(\tau) & \text{if } \tau' > \tau \end{cases}$$

↖ *minus sign* ↗
 when A, B is fermi-like large τ small

The value of τ cannot be arbitrarily large, combining ρ with the factor in $C_j(\tau)$, we have

$$\text{Tr} \left(e^{-\left(\beta - \frac{\tau}{\hbar}\right)(H - \mu N)} C_j e^{-\frac{(\tau - \tau')}{\hbar}(H - \mu N)} C_k^\dagger e^{-\frac{\tau'}{\hbar}(H - \mu N)} \dots \right)$$

if τ is larger than $\hbar\beta$, the factor

$e^{-\left(\beta - \frac{\tau}{\hbar}\right)(H - \mu N)}$ becomes unbounded since H usually has a lower bound but not upper bound.

So, we define the Matsubara Green's function only in the interval $-\hbar\beta \leq \tau, \tau' \leq \hbar\beta$.

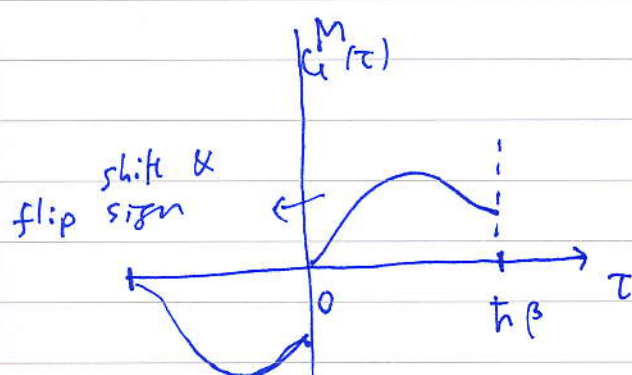
It can be shown that Matsubara function depends only on the difference, $\tau - \tau'$. also satisfies the anti periodic BC. for fermion.

□ Question/problem. prove the above statements explicitly.

We write $G^M(\tau, \tau') \equiv G^M(\tau - \tau')$, and

$$G^M(\tau) = \begin{matrix} \uparrow \\ \text{minus sign for fermion} \\ \text{+ sign for boson} \end{matrix} G^M(\tau - \beta\hbar)$$

[See Piet Brouwer, 2005 lecture notes "Theory of Many-Particle System, as well as Mahan's book]



In particular

$$G^M(\hbar\beta) = -G^M(0)$$

anti-periodic B.C.

Make can make Fourier series (not interval)

$$\tilde{G}^M(i\omega_n) \equiv \int_0^{\hbar\beta} d\tau e^{i\omega_n\tau} G^M(\tau)$$

$$\omega_n = \frac{(2n+1)\pi}{\hbar\beta}$$

$n = 0, \pm 1, \pm 2, \dots$ This is Matsubara frequency

Invers if $G^M(\tau) = \frac{1}{\hbar\beta} \sum_n e^{-i\omega_n\tau} \tilde{G}^M(i\omega_n)$

□ Question: Is $G^M(\tau)$ an odd function of τ ?

ie $G^M(\tau) = -G^M(-\tau)$?

Since $\tau < 0$ part is related to $\tau > 0$ part by anti-symmetry, we focus on $0 \leq \tau \leq \hbar\beta$ only, then

$$G_{AB}^M(\tau) = -\frac{1}{\hbar} \langle A(\tau) B(0) \rangle \quad \begin{array}{l} \text{No need for } T_\tau \\ \text{as } \tau > 0 \end{array}$$

$$0 \leq \tau \leq \hbar\beta$$

$$= -\frac{1}{\hbar} \text{Tr} \left(\frac{e^{-\beta(H-\mu N)}}{Z} e^{\frac{\tau}{\hbar}(H-\mu N)} A e^{-\frac{\tau}{\hbar}(H-\mu N)} B \right)$$

go to Lehmann representation

$$\rightarrow = -\frac{1}{\hbar} \frac{1}{Z} \sum_{n,m} e^{-(\beta - \frac{\tau}{\hbar})(E_n - \mu N_n) - \frac{\tau}{\hbar}(E_m - \mu N_m)} \langle n|A|m\rangle \langle m|B|n\rangle$$

$$\tilde{G}_{AB}^M(i\omega_n) = \int_0^{\hbar\beta} e^{i\omega_n \tau} G_{AB}^M(\tau) d\tau$$

$$= -\frac{1}{\hbar} \frac{1}{Z} \sum_{n,m} \int_0^{\hbar\beta} d\tau e^{\frac{\tau}{\hbar}(E_n - E_m - \mu(N_n - N_m))} e^{+i\omega_n \tau} e^{-\beta(E_n - \mu N_n)} \langle n|A|m\rangle \langle m|B|n\rangle$$

$$= -\frac{1}{\hbar} \frac{1}{Z} \sum_{n,m} \frac{e^{-\beta(E_n - \mu N_n)} \left[e^{\frac{\hbar\beta}{\hbar}(E_n - E_m - \mu(N_n - N_m)) + i\omega_n \hbar\beta} - 1 \right]}{\frac{1}{\hbar}(E_n - E_m - \mu(N_n - N_m)) + i\omega_n} \times \langle n|A|m\rangle \langle m|B|n\rangle$$

$$i\omega_n \hbar\beta = i \left(\frac{2\pi n + 1}{\hbar\beta} \right) \pi \hbar\beta = i2\pi n + i\pi$$

$$e^{i\omega_n \hbar\beta} = -1 \frac{e^{-\beta(E_m - \mu N_m)}}{e^{-\beta(E_n - \mu N_n)}} + e^{-\beta(E_n - \mu N_n)}$$

$$= + \frac{1}{Z} \sum_{n,m} \frac{e^{-\beta(E_n - \mu N_n)}}{E_n - E_m - \mu(N_n - N_m) + i\hbar\omega_n} \langle n|A|m\rangle \langle m|B|n\rangle$$

If we compare this expression with that of Lehmann representation for $\tilde{G}_{AB}^r(E)$ we see that they are the same except the denominator is different.

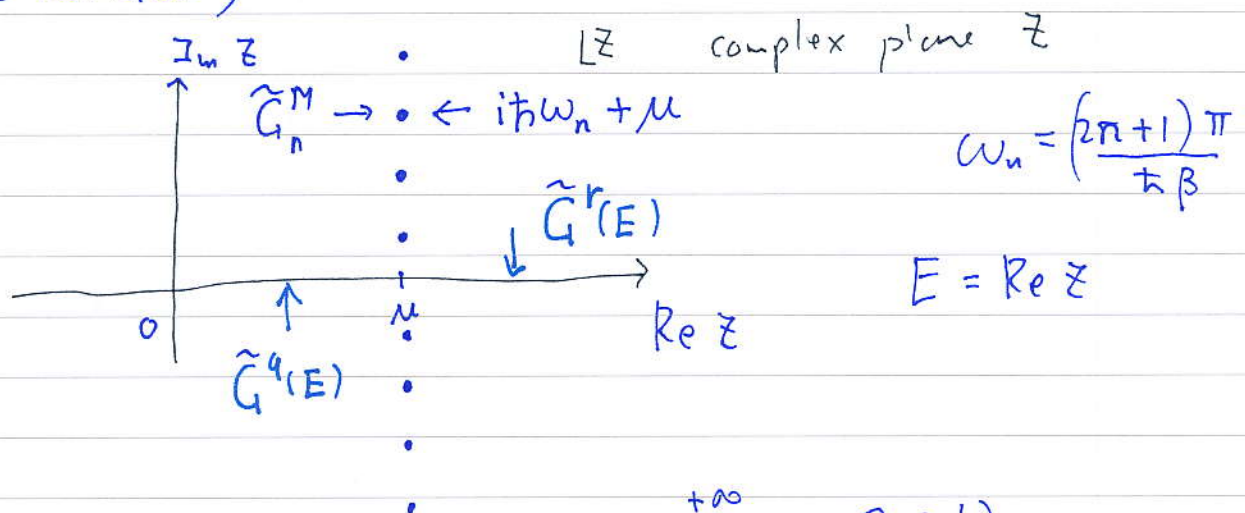
We assume $A = C$, $B = C^\dagger$, so $N_n - N_m = -1$
 (see page 23) so $E + i\eta \leftrightarrow i\hbar\omega_n + \mu$

We have now obtain an important relation between retarded Green's function and Matsubara Green's function. i.e.

$$\tilde{G}^r(E) = \tilde{G}^M(i\hbar\omega_n + \mu \rightarrow E + i\eta)$$

$$\text{or } G(z \rightarrow i\hbar\omega_n + \mu) = \tilde{G}^M(i\omega_n) \equiv \tilde{G}_n^M \quad n=0, \pm 1, \pm 2, \dots$$

picturelly this means

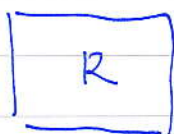
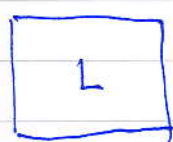


If we already know $G(z) = \int_{-\infty}^{+\infty} \frac{dE'}{2\pi} \frac{\tilde{A}(E')}{z - E'}$

which means we know $\tilde{A}$,
 then if we approach the real axis from above we get retarded Green's function $\tilde{G}^r(E)$, if we approach from below we get $\tilde{G}^a(E)$. if we sit at the discrete points $i\hbar\omega_n + \mu$, where μ is chemical potential of the electron, we are at the values of Matsubara Green's function $\tilde{G}^M(i\omega_n)$. In this sense, $\tilde{A}$ is the most important quantity for equilibrium Green's functions.

After a long preparation for the equilibrium Green's functions, we are ready to talk about the nonequilibrium ones. The reason that we need a thorough understanding of equilibrium one first is that it is always the case that we "build" nonequilibrium Green's functions from equilibrium ones. Without a equilibrium system to start with, we don't know how to make a nonequilibrium one.

The idea of NEGF is that we prepare constrained equilibrium states at $t \rightarrow -\infty$. But separated and noninteracting. at $t \rightarrow -\infty$.



non interacting between L and R

Left system in equilibrium at $\beta_L = \frac{1}{k_B T_L}$, M_L and right system in equilibrium at different β_R , M_R . There's nothing in between. So the Green's function is

$$G_0 = \begin{bmatrix} G_L & 0 \\ 0 & G_R \end{bmatrix}. \quad G_0 \text{ is a matrix, indexed by}$$

positions like j, k or states μ, ν .

We let the system interact as we go to $t \approx 0$, the time of interest.

$$H_{\text{tot}}(t) = H_L + H_R + V e^{\epsilon t}$$

$e^{\epsilon t} \rightarrow 0$ as $t \rightarrow -\infty$, $\epsilon \rightarrow 0^+$. ϵ is known as adiabatic switch-on parameter.

For the electron-photon system, we may consider

$$H_{\text{tot}} = \underbrace{H_L^e + H_R^e}_{\text{free electrons}} + \underbrace{H_\gamma}_{\text{free photons}} + \underbrace{V}_{\text{electron-photon interaction term}}$$

The effect ~~of~~ the interaction is encapsulated in the self-energy.

$$G = G_0 + G_0 \Sigma G \quad \text{or more explicitly}$$

for electrons $\quad \left[\text{this defines the meaning of self-energy } \Sigma \right]$

$$G_{jk}(\tau, \tau') = G_{jk}^0(\tau, \tau') + \int d\tau_1 \sum_{lm} G_{jl}^0(\tau, \tau_1) \Sigma_{lm}(\tau_1, \tau_2) G_{mk}(\tau_2, \tau')$$

so $G_0 \Sigma G$ means matrix multiplication for the site index j, k , and contour integrals for the contour time τ .

For continuous variables like photon, we need integrals for space and contour time, like $\int dx \int d\tau$

The 'NEGF technologies' is discussed in many places including two of my reviews, I don't want to repeat here. In any case, I don't have new insight and cannot make it better than before. So please refer to the literature for this. Just highlight few important points here.

$$G(\tau, \tau') \rightarrow \begin{matrix} + & - \\ \begin{bmatrix} G^{++} & G^{+-} \\ G^{-+} & G^{--} \end{bmatrix} \end{matrix} = \begin{bmatrix} G^t & G^< \\ G^> & G^{\bar{t}} \end{bmatrix}$$

$\tau \equiv (t, \sigma)$

$$\sigma = + \text{ or } -$$

G^t is time order.
 $\bar{t}$ anti-time order

A single G in contour order means 4 G s in the usual time t , $G^t, G^{\bar{t}}, G^>, G^<$. However out of the 4, only two are considered indep't. since we have many linear relations, such as

$$G^t - G^{\bar{t}} = G^r + G^a$$

$$G^> - G^< = G^r - G^a = -iA$$

$$G^r = G^t - G^<, \quad G^a = G^< - G^{\bar{t}}$$

$$G^t + G^{\bar{t}} = G^> + G^<$$

These linear relations can be proved from the basic definition. i.e.

$$G^t(t, t') = \theta(t-t') G^>(t, t') + \theta(t'-t) G^<(t, t')$$

Now ^{for} the Dyson equation on contour

$$G(\tau, \tau') = G^0(\tau, \tau') + \int d\tau_1 \int d\tau_2 G^0(\tau, \tau_1) \underbrace{\Sigma(\tau_1, \tau_2) G(\tau_2, \tau')}_{\text{still matrix multiplication is implied.}}$$

We cannot do Fourier transform

and taking advantage of time translational invariance

which gives us convolution theorem, which is

□ convolution in time t is multiplication in ω or E .

□ Problem: prove the convolution theorem

$$C(t-t') = \int_{-\infty}^{+\infty} A(t-t_1) B(t_1-t') dt_1 \Rightarrow \tilde{C}(\omega) = \tilde{A}(\omega) \tilde{B}(\omega)$$

$$\tilde{A}(\omega) = \int_{-\infty}^{+\infty} dt e^{i\omega t} A(t), \text{ etc.}$$

translational invariance in time, $A(t, t') \equiv A(t-t')$ is needed here.

Go back to real time, the Dyson equation on contour becomes pair of two equations

$$G^r(t, t') = G_0^r(t, t') + \int_{-\infty}^{+\infty} dt_1 \int_{-\infty}^{+\infty} dt_2 G_0^r(t, t_1) \Sigma^r(t_1, t_2) G^r(t_2, t')$$

Applying convolution theorem, we get

$$\tilde{G}^r(E) = \tilde{G}_0^r(E) + \tilde{G}_0^r(E) \tilde{\Sigma}^r(E) \tilde{G}^r(E)$$

This equation is easily solved to give

$$\tilde{G}^r(E) = \left\{ \tilde{G}_0^r(E)^{-1} - \tilde{\Sigma}^r(E) \right\}^{-1}$$

This is a matrix equation including all relevant degrees of freedom (i.e. left and right).

The 2nd equation is the Keldysh equation

$$G^< = G^r \Sigma^< G^a$$

$$\text{or } G^<(t, t') = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} G^r(t, t_1) \Sigma^<(t_1, t_2) G^a(t_2, t') dt_1 dt_2$$

or Apply convolution theorem again

$$\tilde{G}^<(E) = \tilde{G}^r(E) \tilde{\Sigma}^<(E) \tilde{G}^a(E)$$

matrix multiplication implied.

This form of Keldysh equation is valid only in steady state, as we have dropped a transient term. For photon Green's function D , we have similar equations

$$D = D_0 + D_0 \Pi D. \quad \text{How to find } \Sigma \text{ and } \Pi?$$

We need Feynman diagrammatic expansions. I refer to read Fetter & Walecka, or AGD (Abrikosov) or some of the recent ones such as H. Bruus & K. Flensberg.